

Zaffran (2024) PhD Thesis

Post-hoc Predictive Uncertainty Quantification: Methods with Applications to Electricity Price Forecasting

Paper Overview

1. Why this thesis exists

Margaux Zaffran's 2024 PhD thesis Zaffran (2024) synthesises her conformal-prediction research into a single agenda: *post-hoc* predictive uncertainty quantification — prediction intervals that wrap any already-fitted model without retraining it — when the exchangeability assumption fails.

Key Insight

What “post-hoc” means here. You have a trained model. You cannot retrain it — maybe it's expensive, maybe it's owned by someone else, maybe you don't have the training data anymore. Post-hoc UQ wraps an already-fitted model in a procedure that delivers prediction intervals with guarantees. This rules out methods like Bayesian neural networks or ensemble disagreement (which require training-time changes) and rules in the conformal-prediction family.

The applied context is European electricity price forecasting, a domain where the failure modes are concrete and economically consequential. Renewable energy injection makes spot prices volatile; long-term planning requires calibrated uncertainty, not just

point forecasts; the 2021 European energy crisis represented a textbook distribution-shift event that broke conventional forecasting pipelines. Two specific exchangeability failures dominate the setting: *temporal dependence with drift* (the price process is autocorrelated and non-stationary) and *missing covariates* (regulatory data feeds, weather observations, and trader-side inputs are not always available at decision time).

Plain English

Why electricity prices are the right testbed. European spot electricity prices give you both failure modes that the thesis addresses in a single dataset: (i) strong time-series structure (daily/weekly cycles + weather-driven drift), and (ii) genuinely missing covariates (data feeds drop, regulatory updates lag). The 2021 energy crisis adds an unambiguous out-of-sample distribution-shift event to validate against. A method that handles all three is robust to most real-world deployment conditions.

The thesis develops conformal methods for both failures. The ACI / AgACI work Zaffran et al. (2022) addresses the time-series failure. The CP-MDA work Zaffran et al. (2024) addresses the missing-values failure. The thesis assembles these into a unified post-hoc framework and demonstrates it end-to-end on real electricity price data. This overview is a chapter-by-chapter map plus a synthesis of the central theorems.

2. The structure of the thesis

The six chapters trace a methodological progression from problem statement to algorithm to deployment. Chapter 1 (introduction) frames the electricity-market application and the post-hoc UQ research question: given an already-fitted predictor whose construction we cannot influence, what prediction intervals can we wrap around it with guarantees? The framing matters: post-hoc rules out methods (Bayesian neural networks, ensemble disagreement) that require modifying the underlying model.

What This Definition Means

The post-hoc constraint in one sentence. You may not touch the predictor’s training procedure. You may only observe its outputs. The whole CP toolbox respects this constraint by construction; most alternative UQ approaches do not.

Chapter 2 (CP background) introduces the Vovk–Gammerman–Shafer framework Vovk, Gammerman, and Shafer (2005) at the level of detail needed for the rest of the thesis: split conformal, full conformal, the rank-based coverage argument, exchangeability as the minimal assumption, the marginal vs. conditional distinction. The treatment is closer to a textbook chapter than to the Angelopoulos–Bates practitioner tutorial Angelopoulos and Bates (2022) — it gives proofs rather than cookbook recipes.

Chapter 3 (ACI for time series) is the published ACI / AgACI work Zaffran et al. (2022). The chapter analyses Gibbs–Candès ACI Gibbs and Candès (2021) theoretically — showing that the learning rate γ deteriorates efficiency on exchangeable data but improves it on autoregressive data — and introduces AgACI as a parameter-free aggregation that eliminates the γ -tuning problem via online expert aggregation.

Plain English

What Chapter 3 gives you. A complete analysis of online-adaptive CP for time series: how the learning rate trades off responsiveness vs efficiency, why a single γ is never optimal across regimes, and how AgACI sidesteps the choice via expert aggregation. The chapter validates the analysis on French electricity prices including the 2021 distribution shift.

Chapter 4 (missing data and CP) and Chapter 5 (conditional coverage with missing values) together are the published CP-MDA work Zaffran et al. (2024). Chapter 4 establishes that the naive impute-then-predict + conformalise pipeline is marginally valid; Chapter 5 develops mask-conditional validity, proves that CP-MDA-Exact and CP-MDA-Nested achieve it, and shows asymptotic conditional coverage via consistent quantile regression on imputed data.

Plain English

What Chapters 4–5 give you. A full treatment of CP under missing covariates: the marginal-validity free lunch (impute, conformalise, done), the mask-conditional-validity gap (where the free lunch breaks), and the CP-MDA framework that closes the gap. The medical application (TraumaBase) is the validation; the electricity application is where the technique deploys in the broader thesis.

Chapter 6 (applications to electricity forecasting) deploys both ACI/AgACI and CP-MDA on French day-ahead electricity prices, including a case study of the 2021 price explosion. The chapter is where the methodological pieces meet operational constraints: rolling windows, real-time recalibration, latency budgets for the online aggregator.

3. The central guarantees in compact form

The thesis revolves around three theorems and one impossibility result. All four are worth keeping in mind as the through-line.

The marginal coverage baseline. Standard split conformal under exchangeability:

$$\underbrace{\mathbb{P}(Y_{n+1} \in \hat{C}(X_{n+1}))}_{\text{marginal coverage on joint draw of test } (X, Y)} \in \left[\underbrace{1 - \alpha}_{\text{lower bound}}, \underbrace{1 - \alpha + \frac{1}{n+1}}_{\text{upper bound}} \right].$$

This is the result every CP construction in the thesis seeks to preserve or generalise.

What This Result Says

The baseline marginal guarantee. Finite-sample, distribution-free, model-agnostic. Holds under exchangeability of the data, which is satisfied for i.i.d. draws and for any random reshuffling of i.i.d. data. Everything else in the thesis is an attempt to recover something like this guarantee when exchangeability fails (time series, missing data) or to strengthen it (mask-conditional, long-run, conditional).

Long-run coverage under arbitrary sequences (ACI)... The Gibbs–Candès deterministic guarantee that ACI inherits:

$$\underbrace{\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1}\{Y_t \notin \hat{C}_t(X_t)\} - \alpha \right|}_{\text{empirical long-run miscoverage minus nominal level}} \xrightarrow{T \rightarrow \infty} 0 \quad \underbrace{\text{for any sequence } (X_t, Y_t)_{t \geq 1}}_{\text{distribution-free}}.$$

The point: this guarantee requires *no* distributional assumption. ACI converts marginal coverage on exchangeable data into long-run coverage on adversarial sequences — at the price of a wider expected interval width when the data are exchangeable.

What This Result Says

What the ACI guarantee buys. The long-run miscoverage fraction goes to α for *any* sequence, even adversarial. This is a strictly weaker guarantee than the marginal one (which holds at finite n , not just asymptotically) but it holds in settings where the marginal one fails entirely. The trade-off: ACI wastes some interval width on truly exchangeable data, because the algorithm always thinks it might need to adapt.

Mask-conditional validity (CP-MDA)... The stronger coverage notion needed when missing patterns themselves carry information:

$$\underbrace{\mathbb{P}(Y \in \hat{C}_\alpha(X, M) \mid M = m)}_{\text{coverage conditional on specific missing pattern } m} \geq \underbrace{1 - \alpha}_{\text{nominal level}} \quad \underbrace{\forall m \in \mathcal{M}}_{\text{for every possible mask}}.$$

CP-MDA-Exact achieves MCV under MCAR and exchangeability; CP-MDA-Nested achieves MCV under MCAR, exchangeability, and a stochastic-domination assumption on conditional quantiles. The trade-off mirrors the one in CP generally: Exact is statistically tightest but data-hungry (rare pattern matching in high dimensions); Nested uses more calibration data at the cost of slight conservativeness.

What This Result Says

Why MCV is the right thing to demand. Marginal coverage averages over masks; a method satisfying it can still systematically under-cover patients with rare missing patterns. MCV demands coverage *per mask*, which is the fairness property that matters in clinical or actuarial deployments. The CP-MDA framework achieves MCV at finite sample size, which is rare for any conformal extension.

The Lei–Wasserman X-conditional impossibility. The negative result that frames the entire post-hoc enterprise. Distribution-free *X-conditional* coverage — $\mathbb{P}(Y \in \hat{C}(X) \mid X = x) \geq 1 - \alpha$ for every x — is impossible at non-trivial level. Any procedure that achieves it must produce infinite-measure sets on a non-null subset of $\mathcal{X}$. This is why the thesis pursues approximate substitutes (long-run coverage, mask-conditional validity, asymptotic conditional coverage via consistent quantile regression) rather than the unattainable target itself.

Why This Matters

Why the impossibility frames everything. You cannot get per-individual coverage guarantees in the distribution-free post-hoc setting. Full stop. The thesis’s strategy — and the strategy of most of modern CP — is to pick the strongest *achievable* guarantee for each failure mode: long-run for adversarial sequences, mask-conditional for missing data, asymptotic conditional for the consistent-regressor case. None of these are X-conditional in the strict Lei–Wasserman sense, but all are useful substitutes for it in their respective settings.

4. The methodological through-line

A reader who treats ACI and CP-MDA as unrelated will miss the point of the thesis. Both are responses to a structurally similar problem: standard CP assumes exchangeability, the real-world setting violates exchangeability in a specific identifiable way, and the question is what guarantee survives.

For *ACI*, the violation is temporal dependence and drift. The remedy abandons the finite-sample marginal-coverage guarantee in favour of a long-run-average guarantee that survives *any* sequence — a strictly weaker statement, but one that holds where the original cannot.

For *CP-MDA*, the violation is mask-dependent heteroscedasticity — coverage averaged over masks hides catastrophic failure on the rare-mask subpopulations. The remedy restricts calibration to mask-consistent calibration points (Exact) or augments masks systematically (Nested), recovering mask-conditional coverage — a strictly stronger statement than marginal coverage, achievable because the mask is an observed grouping variable.

Key Insight

The thesis’s methodological pattern. Three steps applied repeatedly:

1. Identify the exchangeability failure (time-series, missing data, anything else).
2. Decompose the standard guarantee into the piece that fails and the piece that survives.
3. Engineer a procedure that recovers as much of the surviving piece as possible.

ACI and CP-MDA are two instances of this pattern. The same recipe applies to any new failure mode (multivariate time series, censored data, network data, etc.).

The pattern is the same in both cases: identify the exchangeability failure, decompose the original guarantee into the piece that fails and the piece that survives, then engineer a procedure that recovers as much of the surviving guarantee as possible. The thesis is the canonical example of this approach applied to two different failure modes.

5. Connections and what to read next

The thesis is the bridge between the foundational CP literature (Vovk Vovk, Gamerman, and Shafer (2005), the Lei-Tibshirani-Wasserman CMU lineage) and the applied time-series / missing-data CP literature. For the practitioner-facing companion, read Angelopoulos and Bates Angelopoulos and Bates (2022) alongside Chapter 2. For the time-series cluster, the original ACI paper of Gibbs and Candès Gibbs and Candès (2021) is the predecessor of Chapter 3; the EnbPI line Xu and Xie (2023) is the alternative “residual-refresh” approach to the same problem (Stocker et al. taxonomise both into a four-family picture). For the missing-data cluster, the standard reference for missingness mechanisms is Rubin’s 1976 paper Rubin (1976); the published version of Chapters 4–5 Zaffran et al. (2024) contains the proofs in compact form.

Plain English

The reading order for going further.

1. This overview (you’re reading it).
2. Angelopoulos–Bates 2022 Angelopoulos and Bates (2022) for the practitioner CP recipe.

3. Tibshirani 2023 CMU notes for the statistician’s view (covered separately in this collection).
4. Zaffran 2022 ACI paper Zaffran et al. (2022) for the time-series extension.
5. Zaffran 2024 CP-MDA paper Zaffran et al. (2024) for the missing-data extension.
6. The thesis itself for the unified treatment and the electricity-prices empirical chapter.

The thesis’s particular contribution to the field — beyond the specific algorithms — is the framing of post-hoc UQ as a research programme. Each new domain whose exchangeability fails in a new way (multivariate time series, censored data, network data) becomes a target for the same methodological pattern: diagnose the failure, decompose the guarantee, engineer the procedure. The follow-up work in conformal PID control, multi-distribution-robust CP, and conformal forecasting on functional time series all sits inside this framing, though they were published after the thesis itself.

References

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